

# Homework 5



13.4 PRESS ; unbiased estimator

1) ~~A~~

$$PRESS = \sum_{i=1}^n \frac{\hat{e}_i^2}{(1-h_{ii})^2} = \sum_{i=1}^n \hat{e}_{(i)}^2$$

im diag. score of  $H =$

$$\hat{\sigma}^2 = \frac{RSS}{n-p} = \frac{\sum_{i=1}^n \hat{e}_i^2}{(n-p)} \quad X(X^T X)^{-1} X^T$$

for  $\hat{\sigma}^2 = \frac{PRESS}{\sum_{i=1}^n (1-h_{ii})^2}$  unbiased for  $\sigma^2$  ;

need:  $E(\hat{\sigma}_{PRESS}^2) = \sigma^2$

need:  $E(PRESS) = \sigma^2 \sum_{i=1}^n \frac{1}{1-h_{ii}}$



Under GM;  $Y = X\beta + \varepsilon$   $+ E(\varepsilon) = 0$   
 $Cov(\varepsilon) = \sigma^2 I_n$

residual  $= (I-H)Y = (I-H)\varepsilon$

$Y_i - \hat{Y}_i \sim (0, \sigma^2(I-H))$

$Var(Y_i - \hat{Y}_i) = \sigma^2(1-h_{ii})$

leave one out  $\rightarrow \hat{e}_{(i)} = \frac{Y_i - \hat{Y}_i}{1-h_{ii}}$

$E[\hat{e}_{(i)}^2] = E\left[\frac{e_i^2}{(1-h_{ii})^2}\right] = \frac{1}{(1-h_{ii})^2} E(e_i^2)$

$= \frac{\overset{\sigma^2(1-h_{ii})}{Var(e_i)}}{(1-h_{ii})^2} = \frac{\sigma^2}{1-h_{ii}}$

cont.

$$E[\text{PRESS}] = \sum_{i=1}^n E[\hat{e}(i)^2] = \sum_{i=1}^n \frac{\sigma^2}{1-h_{ii}} = \sigma^2 \sum_{i=1}^n \frac{1}{1-h_{ii}}$$

$$E[\hat{\sigma}_{\text{press}}^2] = \frac{E[\text{PRESS}]}{\sum_{i=1}^n (1-h_{ii})^{-1}} = \frac{\sigma^2 \sum_{i=1}^n (1-h_{ii})^{-1}}{\sum_{i=1}^n (1-h_{ii})^{-1}}$$

$$= \sigma^2$$

✓  
unbiased.

### 14.3 ridge + ols

2)

prove if  $X$  has lin. indep. columns;

$$\begin{aligned}\hat{\beta}^{\text{ridge}}(\lambda) &= (X^T X + \lambda I_p)^{-1} X^T X \hat{\beta} \\ &= V \text{diag}\left(\frac{d_j^2}{d_j^2 + \lambda}\right) V^T \hat{\beta}\end{aligned}$$

where  $\hat{\beta}$  is the OLS coeff.

Ridge:  $\hat{\beta}^{\text{ridge}}(\lambda) = (X^T X + \lambda I_p)^{-1} X^T y$  by defn.

$$\begin{aligned}X &= U D V^T \quad \text{where } U \in \mathbb{R}^{n \times p} \\ D &= \text{diag}(d_1, \dots, d_p) \\ V &\in \mathbb{R}^{p \times p} \text{ orthog.}\end{aligned}$$

$$X^T X = V D^2 V^T$$

$$X^T X + \lambda I_p = V(D^2 + \lambda I_p)V^T$$

$$\begin{aligned}(X^T X + \lambda I_p)^{-1} &= V(D^2 + \lambda I_p)^{-1} V^T \\ &= V \text{diag}\left(\frac{1}{d_j^2 + \lambda}\right) V^T\end{aligned}$$

$$X^T y = V D U^T y$$

$$\Rightarrow \hat{\beta}^{\text{ridge}}(\lambda) = V \text{diag}\left(\frac{1}{d_j^2 + \lambda}\right) V^T \cdot V D U^T y$$

$$= V \text{diag}\left(\frac{d_j}{d_j^2 + \lambda}\right) V^T y$$

↓ const.

Since  $X$  has lin. indep columns;

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

$$= V D^{-2} V^T \cdot V D U^T Y = V D^{-1} U^T Y$$

$$U^T Y = D V^T \hat{\beta}$$

$$\Rightarrow \hat{\beta}_{\text{ridge}}(t) = V \text{diag} \left( \frac{d_j}{d_j^2 + t} \right) D V^T \hat{\beta}$$

$$= V \text{diag} \left( \frac{d_j^2}{d_j^2 + t} \right) V^T \hat{\beta} \quad \checkmark \quad \text{OK}$$

14.4 Ridge as OLS w/ augmented data  
3) prove  $\hat{\beta}_{\text{ridge}}(\lambda) = \text{OLS coeff of } \tilde{Y} + \tilde{X}$

$$\tilde{Y} = \begin{pmatrix} Y \\ 0_p \end{pmatrix} \quad \tilde{X} = \begin{pmatrix} X \\ \sqrt{\lambda} I_p \end{pmatrix}$$

$$\tilde{Y} \in \mathbb{R}^{n+p} \quad \tilde{X} \in \mathbb{R}^{(n+p) \times p}$$

$$\text{OLS}; \quad \tilde{\beta}_{\text{OLS}} = (\tilde{X}^T \tilde{X})^{-1} \tilde{X}^T \tilde{Y}$$

$$\begin{aligned} \tilde{X}^T \tilde{X} &= X^T X + (\sqrt{\lambda} I_p)^T (\sqrt{\lambda} I_p) \\ &= X^T X + \lambda I_p \end{aligned}$$

$$\tilde{X}^T \tilde{Y} = X^T Y + (\sqrt{\lambda} I_p)^T 0_p = X^T Y$$

$$\Rightarrow \tilde{\beta}_{\text{OLS}} = (X^T X + \lambda I_p)^{-1} X^T Y = \hat{\beta}_{\text{ridge}}(\lambda)$$

(ex 3)

15.6 Reduce EN to lasso

4) EN:

$$\operatorname{argmin}_{b \in \mathbb{R}^p} \|y - Xb\|^2 + \lambda \left\{ \alpha \|b\|_2^2 + (1-\alpha) \|b\|_1 \right\}$$

Show  $\nearrow$  reduces to

$$\operatorname{argmin}_{b \in \mathbb{R}^p} \|\tilde{y} - \tilde{X}b\|^2 + \tilde{\lambda} \|b\|_1$$

$$\text{where } \tilde{y} = \begin{pmatrix} y \\ 0_p \end{pmatrix}, \tilde{X} = \begin{pmatrix} X \\ \sqrt{\lambda \alpha} I_p \end{pmatrix}, \tilde{\lambda} = \lambda(1-\alpha)$$

$$\|y - Xb\|^2 + \lambda \alpha \|b\|_2^2 = \|\tilde{y} - \tilde{X}b\|^2$$

~~MAN~~

$$\|\tilde{y} - \tilde{X}b\|^2 = \left\| \begin{pmatrix} y - Xb \\ -\sqrt{\lambda \alpha} b \end{pmatrix} \right\|^2 = \|y - Xb\|^2 + \lambda \alpha \|b\|_2^2$$

plug into EN

$$\|y - Xb\|^2 + \lambda \alpha \|b\|_2^2 + \lambda(1-\alpha) \|b\|_1$$

$$= \|\tilde{y} - \tilde{X}b\|^2 + \tilde{\lambda} \|b\|_1$$

$$\left( \tilde{\lambda} = \lambda(1-\alpha) \right)$$

$$\|\tilde{y} - \tilde{X}b\|^2 + \tilde{\lambda} \|b\|_1 \text{ which is lasso}$$

# hw5

## 5. Lalonde figures

```
library(MatchIt)
library(leaps)
data(lalonde)

y <- lalonde$re78
X_vars <- c("age", "educ", "hisp", "married", "nodegr", "re74", "re75", "treat")

sub_fit <- regsubsets(re78 ~ age + educ + married +
                      nodegree + re74 + re75 + treat,
                      data = lalonde,
                      nvmax = 9,
                      method = "exhaustive")

sub_sum <- summary(sub_fit)
n <- nrow(lalonde)
p_vec <- 1:9 # number of predictors in model of size k

rss_vec <- sub_sum$rss
k_vec <- p_vec + 1 # +1 for intercept

aic_vec <- n * log(rss_vec / n) + 2 * k_vec
```

Warning in  $n * \log(\text{rss\_vec}/n) + 2 * k\_vec$ : longer object length is not a multiple of shorter object length

```
bic_vec <- n * log(rss_vec / n) + k_vec * log(n)
```

Warning in  $n * \log(\text{rss\_vec}/n) + k\_vec * \log(n)$ : longer object length is not a multiple of shorter object length

```
gcv_vec <- (rss_vec / n) / (1 - k_vec / n)^2
```

Warning in  $(\text{rss\_vec}/n)/(1 - k\_vec/n)^2$ : longer object length is not a multiple of shorter object length

```
press_vec <- numeric(9)
for (k in 1:7) {
  vars_k <- names(which(sub_sum$which[k, -1])) # drop intercept column
  formula_k <- as.formula(paste("re78 ~", paste(vars_k, collapse = " + ")))
  fit_k <- lm(formula_k, data = lalonde)
  h_ii <- hatvalues(fit_k)
  e_i <- residuals(fit_k)
  press_vec[k] <- sum((e_i / (1 - h_ii))^2)
}

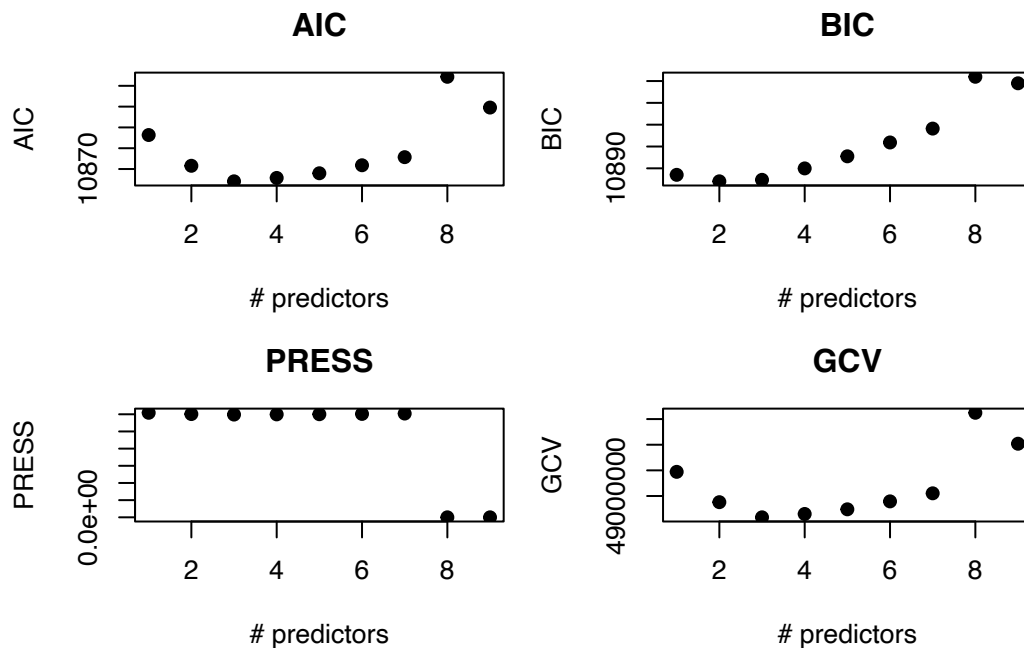
par(mfrow = c(2, 2), mar = c(4, 4, 3, 1))

plot(p_vec, aic_vec, pch = 19,
     xlab = "# predictors", ylab = "AIC", main = "AIC")

plot(p_vec, bic_vec, pch = 19,
     xlab = "# predictors", ylab = "BIC", main = "BIC")

plot(p_vec, press_vec, pch = 19,
     xlab = "# predictors", ylab = "PRESS", main = "PRESS")

plot(p_vec, gcv_vec, pch = 19,
     xlab = "# predictors", ylab = "GCV", main = "GCV")
```



6. Noise , Boston housing data

```
library(MASS)      # Boston data
library(glmnet)    # ridge and lasso
```

Loading required package: Matrix

Loaded glmnet 4.1-10

```
set.seed(42)
data(Boston)

y <- Boston$medv
X_orig <- model.matrix(medv ~ ., data = Boston)[, -1] # 13 original predictors
n <- nrow(X_orig) # 506

# Helper: fit OLS / ridge / lasso and return test MSE
compare_models <- function(X_full, y, label) {
  # 80/20 train-test split
  set.seed(42)
  train_idx <- sample(n, floor(0.8 * n))
```

```

test_idx <- setdiff(1:n, train_idx)

X_tr <- X_full[train_idx, ]; y_tr <- y[train_idx]
X_te <- X_full[test_idx, ]; y_te <- y[test_idx]

# OLS
ols_df <- data.frame(y = y_tr, X_tr)
ols_fit <- lm(y ~ ., data = ols_df)
ols_pred <- predict(ols_fit, newdata = data.frame(X_te))
ols_mse <- mean((y_te - ols_pred)^2)

# Ridge
cv_ridge <- cv.glmnet(X_tr, y_tr, alpha = 0, nfolds = 10)
ridge_pred <- predict(cv_ridge, s = "lambda.min", newx = X_te)
ridge_mse <- mean((y_te - ridge_pred)^2)

# Lasso
cv_lasso <- cv.glmnet(X_tr, y_tr, alpha = 1, nfolds = 10)
lasso_pred <- predict(cv_lasso, s = "lambda.min", newx = X_te)
lasso_mse <- mean((y_te - lasso_pred)^2)

# Number of non-zero lasso coefficients
lasso_nz <- sum(coef(cv_lasso, s = "lambda.min") != 0) - 1 # excl. intercept

cat(sprintf("\n=== %s ===\n", label))
cat(sprintf("  OLS   test MSE: %.3f\n", ols_mse))
cat(sprintf("  Ridge test MSE: %.3f (lambda=%.4f)\n",
            ridge_mse, cv_ridge$lambda.min))
cat(sprintf("  Lasso test MSE: %.3f (lambda=%.4f, non-zero coefs: %d)\n",
            lasso_mse, cv_lasso$lambda.min, lasso_nz))

invisible(list(ols=ols_mse, ridge=ridge_mse, lasso=lasso_mse))
}

res0 <- compare_models(X_orig, y, "Original (p=13)")

```

=== Original (p=13) ===

OLS test MSE: 22.923

Ridge test MSE: 24.217 (lambda=0.6551)

Lasso test MSE: 23.107 (lambda=0.0205, non-zero coefs: 13)

```
noise1 <- matrix(rnorm(n * n), nrow = n)
X1      <- cbind(X_orig, noise1)
res1    <- compare_models(X1, y, "p = n noise columns added (p=519)")
```

Warning in predict.lm(ols\_fit, newdata = data.frame(X\_te)): prediction from rank-deficient fit; attr(\*, "non-estim") has doubtful cases

```
=== p = n noise columns added (p=519) ===
OLS   test MSE: 286267.726
Ridge test MSE: 70.166   (lambda=65.5148)
Lasso test MSE: 32.134   (lambda=0.7359, non-zero coefs: 6)
```

```
noise2 <- matrix(rnorm(n * 2 * n), nrow = n)
X2      <- cbind(X_orig, noise2)          # p = 13 + 1012 = 1025

res2_ridge_lasso <- compare_models(X2[, ], y,
                                   "p = 2n noise columns added (p=1025)")
```

Warning in predict.lm(ols\_fit, newdata = data.frame(X\_te)): prediction from rank-deficient fit; attr(\*, "non-estim") has doubtful cases

```
=== p = 2n noise columns added (p=1025) ===
OLS   test MSE: 286267.726
Ridge test MSE: 73.124   (lambda=65.5148)
Lasso test MSE: 32.120   (lambda=0.7359, non-zero coefs: 9)
```

```
scenarios <- c("Original\n(p=13)", "p = n\n(p=519)", "p = 2n\n(p=1025)")
ols_mse   <- c(res0$ols, res1$ols, NA)    # OLS undefined at p>n_train
ridge_mse <- c(res0$ridge, res1$ridge, res2_ridge_lasso$ridge)
lasso_mse <- c(res0$lasso, res1$lasso, res2_ridge_lasso$lasso)

barplot(rbind(ols_mse, ridge_mse, lasso_mse),
        beside = TRUE,
        names.arg = scenarios,
        col      = c("steelblue", "darkorange", "darkgreen"),
        legend.text = c("OLS", "Ridge", "Lasso"),
        args.legend = list(x = "topleft"),
        ylab      = "Test MSE",
```

```
main = "Boston Housing: OLS vs Ridge vs Lasso\nas noise columns increase",  
ylim = c(0, max(c(ols_mse, ridge_mse, lasso_mse), na.rm=TRUE) * 1.2))
```

